



National University of Computer and Emerging Sciences

FAST School of Computing

CL2005-Database System Lab

Instructor Name: Ahmad Jawad Mustasim

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Office Location/Number: 38 (1st floor F-Block)

Office Hours: Wed & Fri: 9-12 & 3-4, Mon: 3-4, Tue: 9-10:30, Thur: 9-12

Course Information

Program: BS (CS)

Type: Core

Class Meeting Time: 11:30-2:20pm

Credit Hours: 1

Pre-requisites: CS2001 - Data Structures

Class Venue: Lab 14

Course Description/Objectives/Goals:

This course is an introduction to relational databases management Systems. The course will cover fundamental concepts of databases with an emphasis on modeling, designing and implementation of database systems. The theory will be augmented with hands-on exercises on database systems. A project will be conducted in the database system lab that runs in parallel with the course. In the project, the students will develop a data-centric application with a complete set of business transactions and appropriate user interface using a popular programming language and a popular database management system.

Course Learning Outcomes (CLOs):	PLOs
1. Implement a normalized relational design to remove anomalies in a set of relations	(4)
2. Implement the database schema developed against the designed conceptual model	(4)
3. Write queries using relational algebra and SQL	(5)

Lab Weekly Schedule

Lab	Topics	Project Deliverable	Practice/ CLO #
Lab 1	Introduction - Overview of the Outline - Introduction to SQL Server Management studio, Installation & Configuration - Introduction to HTML & CSS Functionality - Practice using Visual Studio Website project	Project Proposal Announcement	2
Lab 2	SQL DDL/DML - Table creation/Insertion / Deletion / Updating Query - Selection Query - Applying Foreign Key/Null/Auto increment constraint Through designer and Query - In Lab Exercise	(i) Project Project Proposal Submission (ii) Deliverable-I Schema Design Announcement	2
Lab 3	SQL Data Retrieval Queries - Select From Where - Ordering/Arithmetic/Operations/Substring/Comparison-Set Operations	Deliverable-I Queries/Discussion	3
Lab 4	SQL Joins & Aggregation - Aggregation. Group By, Order By - Joins & types (inner, outer, cross) - Node JS Intro & Db Connection	(i) Deliverable-I Submission (ii) Deliverable II (Queries For Features) Announcement	3
Lab 5	SQL Nested Queries - Advanced SQL (Nested and Subquery) - Node JS Practice	Deliverable II Queries/Discussion	3
Lab 6	Stored Procedures & Views, UI - Stored Procedures - Views (Virtual Tables) - Practice with HTML, CSS	Deliverable II Queries/Discussion	3
Lab 7	CRUD Operations from UI - Basic UI Page with any JS frontend: ReactJS or VueJS or Angular JS - Login Form and CRUD Operations on a Table	(i) Deliverable II Submission (ii) Deliverable III: Minimal UI Announcement	

Lab 08	Mid Term	-----	2,3
Lab 09	Triggers and Types of Triggers - Practice with Triggers - Schema Normalization Exercises	(i) Deliverable III: Minimal UI Submission (ii) Deliverable IV: Schema Normalization & UI Update	3
Lab 10	Introduction to ERWIN & DB Connectivity - Creating Relational model using ERWIN	Deliverable IV Queries/Discussion	2
Lab 11	Transactions & Advance Concepts	Deliverable IV Submission	3
Lab 12	Complete Project Evaluation	Viva	

Tentative Grading Criteria:

1. In Lab + Post Labs (25%)
2. Mid Term (20%)
3. Project (15%)
 - a. Deliverable I: Schema Design 3%
 - b. Deliverable II: Queries For Features 4%
 - c. Deliverable III: Features with Minimal UI 4%
 - d. Deliverable IV: Normalized Schema and UI Update 4%
4. Final (40%)

Lab Policies:

1. The lab outline is tentative.
2. At least **80% attendance** is compulsory.
3. Absolute Grading will be followed and the minimum requirement to pass this lab is to obtain at least 50% marks.
4. University's rules about code of conduct apply in the lab as well.
5. Seating Plan is mandatory to be followed and can be shared for any assessment i.e. labs, exams, quizzes etc
6. Plagiarism in any work (Labs, Quiz, Assignment, Midterms, and Final Exam) from any source (Internet or a Student) will result in an **F grade**.

Note: Lab manuals will be available through Google Classroom or Xeon.